

ITAI-1371 - Intro to Machine Learning
Individual Contribution Journal

Camila Ferreira da Silva

My contribution to Lab 03 included working through the notebook, reviewing its results, developing an education application, and coordinating the group's documentation. I also took responsibility for bringing the materials together and preparing the final submission.

I executed the notebook's code cells and reviewed the stages of the machine learning workflow, including data exploration, feature selection, training and testing, and model evaluation. I examined the dataset information and visualizations to understand how these steps supported the classification task.

One area I focused on was interpreting the classification report and confusion matrix. Precision and recall were challenging for me, so I reviewed them alongside the model's predictions to understand the difference. I used that understanding to explain why overall accuracy does not describe every aspect of model performance.

For the real-world application, I selected Education and connected the lab to my experience as a teacher. I proposed using supervised learning to identify students who may need additional support. I described the problem, the data that would be needed, and the importance of teacher oversight, privacy, and fairness.

Within the group, I helped organize the work and reviewed classmates' documents. I gathered their reflection and contribution materials, organized the documents, and checked their formatting. I also handled the organization of the GitHub materials and the preparation of PDF files for submission.

Great Inioluwa Olayiwola

In this practical machine learning lab, I built a full classification pipeline using the Scikit-Learn Wine dataset. First, I got the setup ready. Then I looked at the data with basic Exploratory Data Analysis. I also used heatmap plots to see patterns. Next, I split the dataset into training and test parts. I used stratification so the class mix stayed similar in both sets. After that, I trained two models: Logistic Regression and a Decision Tree. I checked how well they worked using accuracy scores. I also reviewed classification reports and confusion matrices. To learn more, I tried different feature choices. For example, I focused on alcohol, color_intensity, and proline to see how each one affected prediction strength. At the end, I answered a concept question about learning types. I sorted examples into supervised, unsupervised, and reinforcement learning.

Karima Sadruddin

1. Notebook and code

My specific contribution:

My specific contribution was working on the Entertainment section of the project, using Netflix as an example. I focused on understanding how machine learning can be applied in the entertainment industry, particularly for analyzing content and making recommendations based on user interests and viewing behavior.

Cells or workflow stages I worked on:

I worked on the section discussing machine learning applications in Entertainment and connected the example to the overall machine learning workflow discussed in the notebook.

Code I ran, checked, or changed:

I reviewed the notebook examples and focused on understanding how machine learning concepts could be applied to an entertainment platform such as Netflix. I did not make changes to the main Wine classification model unless specifically assigned by the team.

2. Data exploration and model evaluation

My contribution to exploring the Wine dataset, reviewing features, or checking data quality:

My main contribution was not the Wine dataset analysis. I reviewed the notebook to understand how the dataset, features, training/testing process, and model evaluation fit into the overall project so I could connect the concepts to my Entertainment/Netflix section.

Models or metrics I examined and what I learned:

I reviewed the machine learning workflow, including model training and evaluation. The notebook compared Logistic Regression and Decision Tree models using accuracy and classification reports. I learned that machine learning models can be evaluated using metrics such as accuracy, precision, recall, and F1-score.

Any issue I investigated and how I addressed it:

I focused on understanding how the general machine learning workflow could be applied to an entertainment example. I reviewed the notebook material and connected the concepts to Netflix and entertainment-related applications.

3. Team coordination and communication

What I communicated, reviewed, explained, or shared with teammates:

I shared information and ideas related to the Entertainment/Netflix application of machine learning and reviewed how this section connected with the other parts of the project.

Any feedback I gave or incorporated:

I reviewed the project requirements and incorporated feedback from the team to make sure my Entertainment/Netflix contribution was connected to the machine learning concepts covered in the notebook.

4. Evidence and reflection

Evidence of my work:

My contribution is reflected in the Entertainment/Netflix section of the project and the related notes/materials shared with the team.

What I personally learned from completing my part:

I learned how machine learning can be applied beyond the Wine classification example and can be used in the entertainment industry. Using Netflix as an example helped me understand how data about users and content can be used to identify patterns and support personalized experiences.

What I would contribute differently next time:

Next time, I would document my contribution more clearly as I work, save screenshots or notes of the specific section I complete, and communicate my progress with the team earlier.

Quan Thach

Transportation: In transportation, ML powers autonomous vehicles by combining computer vision and sensor data to detect lanes, pedestrians, and traffic signs, then using learned driving policies (often trained with supervised and reinforcement learning) to steer, brake, and navigate. It also drives route optimization: models trained on historical and live traffic data predict congestion and travel times, letting apps like Google Maps or delivery fleets choose faster routes and cut fuel costs. Other applications include predictive maintenance for trains and aircraft (spotting failures before they happen), demand forecasting for ride-sharing and public transit, and dynamic pricing for tickets and rides.